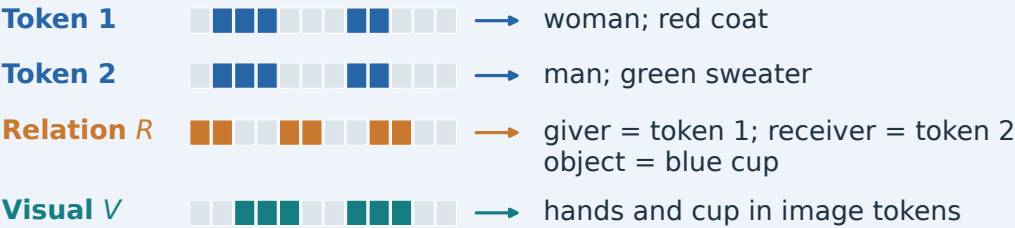


Learn representation, generation, and computation together

REPRESENTATION: token vectors and component extractors



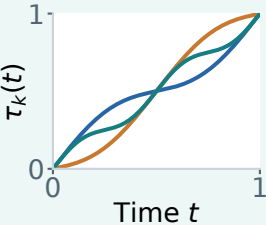
Colored cells illustrate selected coordinates; useful components are learned.

Make informative components ready earlier to improve generation

Vary noise to learn which components help predict other components

Choose component inputs and targets; share attention across tasks; specialize prediction modules.

DENOISING ALGORITHM



Each component s_k has its own schedule. Schedules overlap; order can change.

TRANSFORMER ARCHITECTURE

Conditions



Each task also receives its noisy target.

$\uparrow$ Jointly train schedules and transformer $\downarrow$